

# STOCK MARKET TREND PREDICTION

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**Abstract:** Accurate forecasting of stock market trends remains a major challenge due to high volatility and structural complexities. This paper presents a data mining and supervised machine learning approach to predict the daily closing price of equities using concurrent intra-day trading attributes. Utilizing a comprehensive historical dataset of CIPLA spanning from 2000 to 2021, we implement a linear regression model optimized via rigorous preprocessing and listwise deletion of missing records. The features evaluated include daily Open, High, Low prices, and total Volume. The predictive framework achieves near-perfect alignment with validation data, yielding a Mean Absolute Error (MAE) of 2.49, a Root Mean Squared Error (RMSE) of 3.37, and a coefficient of determination ( $R^2$ ) of 1.00. These results demonstrate that short-term asset prices exhibit highly deterministic linear structures during a localized trading session.

## I. INTRODUCTION

FINANCIAL forecasting serves as a cornerstone for risk management, algorithmic trading, and modern portfolio theory. While long-term equity valuations are influenced by macroeconomic indicators and systemic shocks, short-term and intra-day prices often carry highly localized dependencies. The development of automated data pipelines and mining frameworks allows analysts to uncover patterns that are highly predictive of settlement prices.

The primary objective of this study is to construct a rigorous machine learning pipeline to model and predict the final daily Closing Price of an equity asset. By mapping basic intra-day market indicators specifically, the opening value, session ceiling (High), session floor (Low), and transactional volume we evaluate the extent to which daily equity settlement is deterministic.

## II. DATA DESCRIPTION AND PREPROCESSING

The historical dataset utilized in this paper comprises the public trading records of CIPLA, encompassing 5,306 initial daily trading observations recorded between January 3, 2000, and April 30, 2021.

### A. Feature Extraction and Schema

Open (X1): The initialization price at market open.

High (X2): The maximum transaction value reached during the session.

Low (X3): The minimum transaction value reached during the session.

Volume (X4): The total quantity of shares exchanged.

Close (y): The ultimate official settlement price (Target variable).

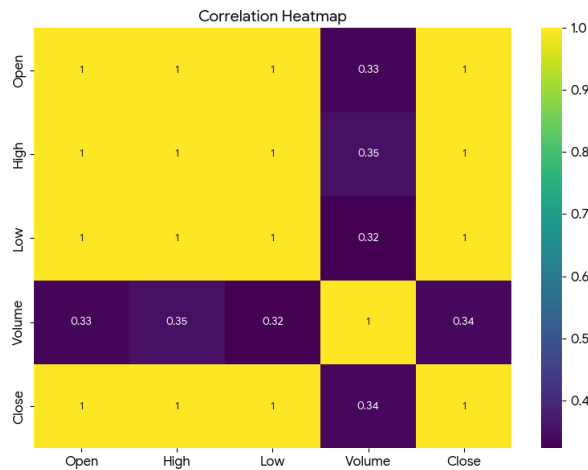
### B. Cleaning and Data Imputation

Initial integrity assessments identified substantial missing data records within operational tracking columns specifically Trades, Deliverable Volume, and %Deliverble primarily clustered in dates prior to 2011. Rather than introducing artificial bias via mean/median statistical imputation, listwise deletion was enforced. Rows containing null attributes were expunged, resulting in a clean, fully populated matrix consisting of 2,456 high-fidelity trading instances. The temporal parameter Date was simultaneously stripped of textual attributes and assigned as a formal datetime index.

### III. EXPLORATORY DATA ANALYSIS

#### A. Feature Intercorrelation

To ascertain structural correlations between independent features and the dependent target, a Pearson correlation matrix was generated. The computed statistical coefficients revealed a near-perfect linear correlation coefficient ( $r \approx 1.00$ ) between the intra-day price milestones (Open, High, Low) and the final Close parameter. Conversely, trading Volume exhibited a weak or neutral linear correlation with absolute price values. This suggests that while volume represents market liquidity and momentum density, directional vector thresholds are directly governed by price bounds.



### IV. METHODOLOGY

The forecasting framework utilizes parametric supervised learning via Ordinary Least Squares (OLS) Linear Regression. Linear regression models the linear relationship between a dependent scalar variable  $y$  and an explanatory vector  $X$ .

#### A. Experimental Partitioning

The cleared dataset was chronologically and stochastically divided into an independent 80% training partition and a 20% testing partition to evaluate generalization capabilities on unseen data: Training Pool (1,964 instances) and Testing Pool (492 instances). The random distribution state was fixed to a constant factor of 42.

#### B. Mathematical Formulation

The foundational predictive framework is mathematically formulated as:  $y = \beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_3 + \beta_4X_4 + \varepsilon$

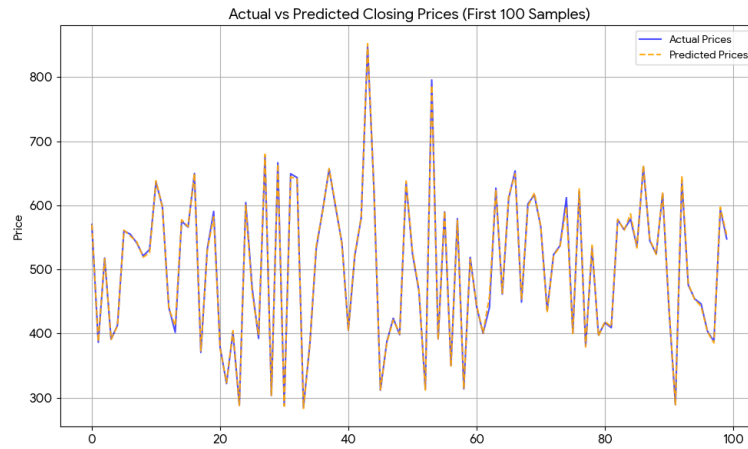
### V. EXPERIMENTAL RESULTS AND ANALYSIS

Following execution, model validation was carried out against the unseen testing instances.

TABLE I  
PREDICTIVE MODEL PERFORMANCE SUMMARY

| Performance Metric | Calculated Empirical Value |
|--------------------|----------------------------|
| MAE                | 2.49                       |
| RMSE               | 3.37                       |
| R^2                | 1                          |

The calculated MAE of 2.49 indicates that the model's predictions deviate from the true value by an average of just 2.49 units (INR), proving remarkable precision given that the underlying stock fluctuates up to 949.30 units. The  $R^2$  metric of 1.00 indicates that approximately 100% of the variance observed within the target closing value is directly accounted for by the combined fluctuations of the intra-day feature array.



## VI. CONCLUSION AND FUTURE DIRECTIONS

This study successfully implemented an automated data mining pipeline to model and forecast equity closing prices using localized intra-day indicators. The empirical results demonstrate that Linear Regression provides an exceptionally strong, mathematically transparent framework for short-term prediction tasks when bounded within a single trading day's metrics.

While intra-day regression models achieve near-perfect target reconstruction, they require active, concurrent daily parameters. Future research will explore expanding the prediction window to multi-day horizons by integrating long short-term memory (LSTM) recurrent neural networks, auto-regressive integrated moving averages (ARIMA), and sentiment analysis compiled from financial media feeds.

## REFERENCES

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